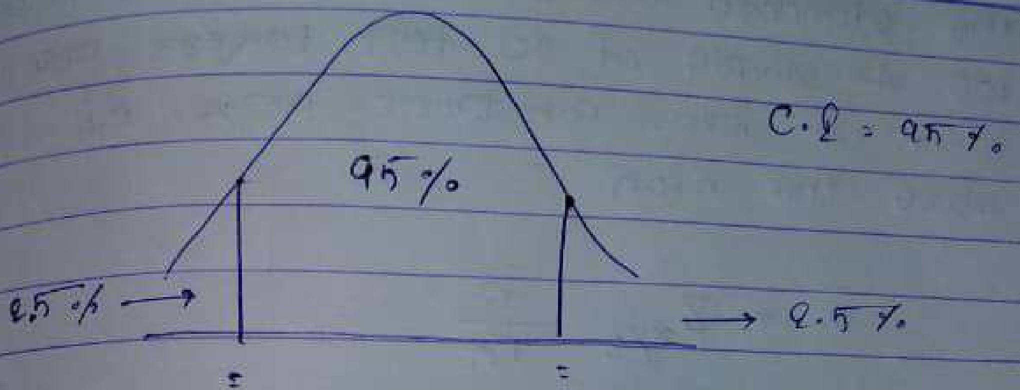
Confidence Intervals and Margin of Error



Point Estimate

$\boxed{\bar{x}} \longrightarrow \boxed{\mu}$

$\bar{x} = 2.5$ $\mu = 3$

Confidence Interval

Point Estimate ± Margin of Error

z test $\Rightarrow \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

∴ If t test then use t in place of z.

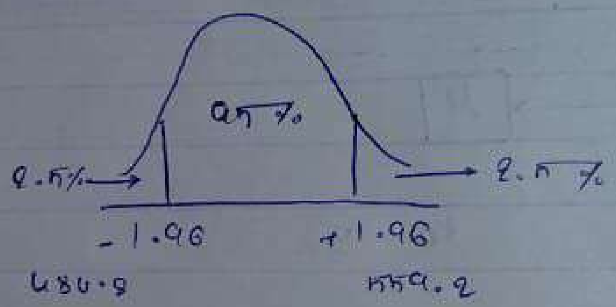
eg:- On the verbal section of CAT exam, the standard deviation is known to be 100. A sample of 90 test takers has a mean of 520. Construct 95% C.I above the mean.

$$\bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$\alpha = 0.05$

$1 - 0.025 = 0.9750$

$+1.96 \Rightarrow Z$ table



$$\text{lower C.I} = 520 - (1.96)\frac{100}{\sqrt{25}} = 480.8$$

$$\text{Higher C.I} : 520 + (1.96)\frac{100}{\sqrt{25}} = 559.2$$

Conclusion :

I am 95% confident about the mean CAT score is between 480.8 and 559.2